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APPARATUSES AND METHODS FOR FACILITATING A GENERATION AND UTILIZATION OF NETWORKS ASSOCIATED WITH EVENTS

Abstract

Aspects of the subject disclosure may include, for example, obtaining first traffic from a first communication device; analyzing the first traffic to classify the first traffic as being associated with an execution of a first application; based on the classification of the first traffic as being associated with the execution of the first application, translating an address associated with the first traffic from a first address associated with a first network to a second address associated with a second network, the second network being different from the first network; and conveying the first traffic to a second communication device of the second network using the second address. Other embodiments are disclosed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. patent application Ser. No. 17/557,107, filed on Dec. 21, 2021. All sections of the aforementioned application(s) are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE
[0002] The subject disclosure relates to apparatuses and methods for facilitating a generation and utilization of networks associated with events.
BACKGROUND
[0003] As the world becomes increasingly connected through vast communication networks and via various communication devices, additional opportunities are created/generated to convey data from a first communication device to one or more other communication devices. Such data may pertain to one or more programs, applications, or the like. One of the fastest growing entertainment sectors is in the area of electronic sports (also referred to as eSports in the art). eSports may entail individuals or teams competing against one another, both live and on the web/online, in relation to a playing of a game (e.g., a video game). Some eSports competitions feature incentives (e.g., prizes, sponsorships, etc.) that are awarded to victorious individuals or teams, which tends to raise the profile and competitiveness of the games that are played.
[0004] In many instances (e.g., in peer-to-peer gaming scenarios), an Internet Protocol (IP) address and port configuration of a client device (e.g., a personal computer (PC), a mobile device, a tablet, a console, etc.) of a user/gamer is, by design, communicated and exposed to the other users/gamers in a multiplayer competition. This enables nefarious users/gamers to direct attacks, such as a distributed denial-of-service (DDOS) attack and/or malicious traffic, at the IP address of the user/gamer, granting them an unfair edge during competitions. Still further, nefarious users/gamers (or others) may correlate/formulate a relationship between the IP address and other online activity, including activity on forums, in voice/video chat application, etc. Given that IP addresses tend to be associated with geographic areas/regions, even more information regarding an individual user/gamer may be exposed. Taken collectively, the information may open the individual user/gamer up to side-channel attacks, up-to and including “SWATing”. SWATing entails the calling in of a false, but serious, allegation of imminent domestic violence to authorities (e.g., a police department), attempting to cause the authorities to send a special weapons and tactics (SWAT) team (or the like) to the target's location (e.g., residence). As one skilled in the art will appreciate, SWATing imposes great danger not only on the individual user/gamer, but on everyone involved, inclusive of the authorities and innocent bystanders.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0006] FIG. **1** is a block diagram illustrating an exemplary, non-limiting embodiment of a communications network in accordance with various aspects described herein.
[0007] FIG. **2**A is a block diagram illustrating an example, non-limiting embodiment of a system functioning within the communication network of FIG. **1** in accordance with various aspects described herein.
[0008] FIG. **2**B depicts an illustrative embodiment of a method in accordance with various aspects described herein.
[0009] FIG. **3** is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.
[0010] FIG. **4** is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.
[0011] FIG. **5** is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.
[0012] FIG. **6** is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

[0013] The subject disclosure describes, among other things, illustrative embodiments for routing or conveying traffic/data in networks in accordance with an identification of an application associated with the traffic/data. Other embodiments are described in the subject disclosure.
[0014] One or more aspects of the subject disclosure include, in whole or in part, monitoring first traffic from a first device, wherein the first device has a first network address associated with a first network, classifying, based on the monitoring of the first traffic, the first traffic as first gaming traffic, based on the classifying of the first traffic as first gaming traffic, establishing a second network that is different from the first network, mapping the first network address to a second network address that is different from the first network address, wherein the second network address is associated with the second network, and routing the first traffic to a gaming server of the second network using the second network address.
[0015] One or more aspects of the subject disclosure include, in whole or in part, obtaining first traffic from a first communication device, analyzing the first traffic to classify the first traffic as being associated with an execution of a first application, based on the classification of the first traffic as being associated with the execution of the first application, translating an address associated with the first traffic from a first address associated with a first network to a second address associated with a second network, the second network being different from the first network, and conveying the first traffic to a second communication device of the second network using the second address.
[0016] One or more aspects of the subject disclosure include, in whole or in part, analyzing first data to identify that the first data is associated with an online game and to identify that the first data is associated with a first device, the first device having a first address associated with a first network, establishing, based on the analyzing of the first data, a second network that is different from the first network, translating the first address to a second address that is different from the first address, wherein the second address is associated with the second network, and transmitting the first data to a communication device in the second network using the second address.
[0017] Referring now to FIG. **1**, a block diagram is shown illustrating an example, non-limiting embodiment of a system **100** in accordance with various aspects described herein. For example, system **100** can facilitate in whole or in part monitoring first traffic from a first device, wherein the first device has a first network address associated with a first network, classifying, based on the monitoring of the first traffic, the first traffic as first gaming traffic, based on the classifying of the first traffic as first gaming traffic, establishing a second network that is different from the first network, mapping the first network address to a second network address that is different from the first network address, wherein the second network address is associated with the second network,

and routing the first traffic to a gaming server of the second network using the second network address. System **100** can facilitate in whole or in part obtaining first traffic from a first communication device, analyzing the first traffic to classify the first traffic as being associated with an execution of a first application, based on the classification of the first traffic as being associated with the execution of the first application, translating an address associated with the first traffic from a first address associated with a first network to a second address associated with a second network, the second network being different from the first network, and conveying the first traffic to a second communication device of the second network using the second address. System **100** can facilitate in whole or in part analyzing first data to identify that the first data is associated with an online game and to identify that the first data is associated with a first device, the first device having a first address associated with a first network, establishing, based on the analyzing of the first data, a second network that is different from the first network, translating the first address to a second address that is different from the first address, wherein the second address is associated with the second network, and transmitting the first data to a communication device in the second network using the second address.

[0018] In particular, in FIG. **1** a communications network **125** is presented for providing broadband access **110** to a plurality of data terminals **114** via access terminal **112**, wireless access **120** to a plurality of mobile devices **124** and vehicle **126** via base station or access point **122**, voice access **130** to a plurality of telephony devices **134**, via switching device **132** and/or media access **140** to a plurality of audio/video display devices **144** via media terminal **142**. In addition, communication network **125** is coupled to one or more content sources **175** of audio, video, graphics, text and/or other media. While broadband access **110**, wireless access **120**, voice access **130** and media access **140** are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices **124** can receive media content via media terminal **142**, data terminal **114** can be provided voice access via switching device **132**, and so on).

[0019] The communications network **125** includes a plurality of network elements (NE) **150**, **152**, **154**, **156**, etc. for facilitating the broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or the distribution of content from content sources **175**. The communications network **125** can include a circuit switched or packet switched network, a voice over Internet protocol (VOIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

[0020] In various embodiments, the access terminal **112** can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals **114** can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

[0021] In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

[0022] In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VoIP telephones and/or other telephony devices.

[0023] In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal **142**. The display devices **144** can include televisions with or without a set top box, personal computers and/or other display devices.

[0024] In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

[0025] In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150**, **152**, **154**, **156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

[0026] FIG. **2**A is a block diagram illustrating an example, non-limiting embodiment of a system **200***a* in accordance with various aspects described herein. In some embodiments, the system **200***a* may function within, or may be operatively overlaid upon, the communication network **100** of FIG. **1**. The system **200***a* may be operative in conjunction with one or more programs, applications, or the like. In some embodiments, the system **200***a* may be utilized to facilitate a competition, such as for example an online game or an eSports competition played between two or more users or players.

[0027] The system **200***a* may include a number of communication devices. For example, the system **200***a* may include user equipment/client devices **202***a***-1**, **202***a***-2**, **202***a***-3**, and **202***a***-4**. In general, a client device may utilize wired connectivity, wireless connectivity, or a combination thereof, in respect of accessing/obtaining/facilitating communication services. In some embodiments, a client device may include (without limitation): a mobile device, a personal computer (PC), a laptop computer, a tablet, a gaming console, etc.

[0028] In some embodiments, the system **200***a* may be utilized to facilitate a competition, such as an eSports competition. In this respect, the client devices **202***a***-1** through **202***a***-4** may be arranged as two or more teams as part of the competition. To demonstrate, a first team may include the client device **202***a***-1** and a second team may include the client devices **202***a***-2** through **202***a***-4**. As this example demonstrates, there is no requirement that the teams include a same number of client devices (e.g., in this particular example the first team includes one client device and the second team includes three client devices). Of course, in some embodiments a same number of client devices (or, analogously, a same number of users/gamers) in respect of two or more teams may be utilized, which may promote/encourage fairness as between the teams in the competition.

[0029] In some embodiments, a conveyance of traffic/data associated with an execution of an application (e.g., a game) may be handled/facilitated by a gateway, a router, or the like, illustratively depicted via reference character **206***a* in FIG. **2**A. For example, and assuming that the client devices **202***a***-2** through **202***a***-4** are located at a residence/residential location, the device **206***a* may be referred to as a residential gateway (RG).

[0030] The gateway **206***a* may be communicatively coupled to one or more servers, such as gaming servers **210***a*. The gaming servers **210***a* may host data associated with a game, potentially as part of a competition. In some embodiments, a communication link or channel between the gateway **206***a* and the gaming servers **210***a* may be used to authenticate one or more of the client devices **202***a***-2** through **202***a***-4**; the communication link/channel may be secure. The client device **202***a***-1** may be authenticated in a similar manner.

[0031] As described above, conventionally the participation of a user in a competition or game may expose information associated with the user (or an associated client device) to other users. For example, and as described above, user participation in the game may tend to expose an IP address (or other identifier) of a client device of the user to other users or devices. To reduce (e.g., avoid) such exposure, a secondary/separate network may be established to host at least part of the game or competition. The secondary network may include one or more of: the gateway **206***a*, gaming servers **210***a*, and domain name system (DNS)/network address translation (NAT) devices, such as DNS/NAT devices **218***a***-1** and **218***a***-2**. In some embodiments, functionality associated with the gaming servers **210***a*, the DNS/NAT device **218***a***-1**, and/or the DNS/NAT device **218***a***-2** may be implemented via hardware, software, firmware, or any combination thereof.

[0032] In practice, the secondary network may host secondary traffic/data associated with an application, such as a game or a competition. This may be contrasted with other traffic/data, such as typical or primary traffic/data associated with a primary network. For example, the primary network may include Internet-hosted peer devices and/or network infrastructure, as generally represented by reference characters **222***a* and **226***a* in FIG. **2**A. In some embodiments, an Internet-hosted peer **226***a* may participate in a gaming session or competition. For example, an Internet-hosted peer **226***a* may include/utilize an over-the-top (OTT) virtual private network (VPN) technology/technique to participate in the gaming session or competition.

[0033] The DNS/NAT device **218***a***-1** (which, in some embodiments, may be hosted on/at the gateway **206***a*) may classify traffic/data associated with the client devices **202***a***-2** through **202***a***-4** as being primary traffic (e.g., typical traffic associated with the Internet) or secondary traffic (e.g., traffic associated with a game or competition). Based on that classification, and assuming that the traffic in question is secondary traffic, the DNS/NAT device **218***a***-1** may route the traffic through the secondary network (e.g., may route the traffic to/from the gaming servers **210***a*, potentially by way of another device, entity or network generally represented by reference character **234***a* in FIG. **2**A). In some embodiments, the DNS/NAT device **218***a***-1** may incorporate/include a rapidly-rotating, geographically-independent DNS/NAT pool of addresses that prevents nefarious actors from identifying a location of the gateway **206***a* or a location of the client devices **202***a***-2** through **202***a***-4**, or associating activity from one session to another. In this manner, a respective location-dependent primary (e.g., IP) address associated with the devices **202***a***-2** through **202***a***-4** and the gateway **206***a* may be shielded/obscured from view via the use of a secondary address facilitated by the DNS/NAT device **218***a***-1**.

[0034] At least some of the functionality described above in respect of the DNS/NAT device **218***a***-1** may be included/incorporated as part of the DNS/NAT device **218***a***-2** in conjunction with traffic/data associated with the client device **202***a***-1**. For example, the DNS/NAT device **218***a***-2** may classify traffic/data associated with the client device **202***a***-1** as being primary traffic (e.g., typical traffic associated with the Internet) or secondary traffic (e.g., traffic associated with a game or competition). Based on that classification, and assuming that the traffic in question is secondary traffic, the DNS/NAT device **218***a***-2** may route the traffic through the secondary network (e.g., may route the traffic to/from the gaming servers **210***a*, potentially by way of device **234***a* in FIG. **2**A). In some embodiments, the DNS/NAT device **218***a***-2** may incorporate/include a rapidly-rotating, geographically-independent DNS/NAT pool of addresses that prevents nefarious actors from identifying a location of the client device **202***a***-1**, or associating activity from one session to another. In this manner, a location-dependent primary (e.g., IP) address associated with the device **202***a***-1** may be shielded/obscured from view via the use of a secondary address facilitated by the DNS/NAT device **218***a***-2**. In some embodiments, (functionality associated with) the DNS/NAT device **218***a***-2** may be at least partially hosted at/by the client device **202***a***-1**.

[0035] In some embodiments, DDOS functionality (which may be hosted at/by the DNS/NAT device **218***a***-1**, the DNS/NAT device **218***a***-2**, the device **234***a*, or any combination thereof), potentially in combination with an authentication or verification of client or user device identity, may reduce (e.g., eliminate) an ability of a nefarious actor/user to utilize network traffic to gain an unfair advantage. Via the use of the secondary network described above, traffic/data routing can be enhanced (e.g., optimized) by reducing a number of hops. In this way, traffic/data may be provided to game servers or client devices as quickly and efficiently as possible, thereby improving the gaming experience.

[0036] Referring now to FIG. **2**B, an illustrative embodiment of a method **200***b* in accordance with various aspects described herein is shown. The method **200***b* may be implemented (e.g., executed), in whole or in part, in conjunction with one or more systems, devices, and/or components, such as for example the systems, devices, and components described herein. The method **200***b* may facilitate a performance of one or more operations, described below in relation to the blocks shown in FIG. **2**B. The method **200***b* may be implemented to facilitate a competition that may take place at one or more locations, potentially in a connected or online environment/context.

[0037] In block **204***b*, traffic/communications associated with one or more communication devices (e.g., one or more client devices, one or more gateways, one or more servers, etc.) may be monitored. For example, as part of block **204***b* the traffic may be sampled via a network probe, a monitoring sensor, or the like. In some embodiments, block **204***b* may include a mobile edge computing (MEC) device that is proximal to the location of the communication devices monitoring the traffic.

[0038] The monitoring of block **204***b* may be based on, or include, an analysis of characteristics/parameters of the traffic. For example, in some embodiments the monitoring of block **204***b* may be based on an examination/analysis of headers, metadata, or the like, associated with the traffic. In some embodiments, the monitoring might not include access to the contents/payload of the traffic, which can be useful in environments where (user) privacy and security are important. For example, the monitoring may include an analysis of patterns in the traffic (potentially relative to other traffic) without an examination of the specifics of the payload of the traffic in question. In some embodiments, a monitoring of traffic (or various aspects associated therewith) may be performed subject to authorization or approval of participants/users, including via opt-in and/or opt-out techniques.

[0039] Based on the monitoring of block **204***b*, the traffic may be classified in block **208***b*. For example, block **208***b* may include classifying the traffic as being associated with an application (e.g., a game or a competition) or not being associated with the application.

[0040] In block **212***b*, a determination of a routing of the traffic may be generated. The determination of block **212***b* may be based on the classification of block **208***b*. For example, if the traffic is classified as being associated with the application in block **208***b*, a secondary network (potentially including a gaming/game server) may be selected for routing the traffic as part of block **212***b*. On the other hand, if the traffic is classified as not being associated with the application in block **208***b* (e.g., the traffic pertains to other traffic, such as typical, Internet-bound traffic), a primary network (potentially excluding or omitting a gaming/game server) may be selected for routing the traffic as part of block **212***b*.

[0041] In block **216***b*, a selection of an address to be used for routing the traffic may be provided for. For example, if as part of block **212***b* a determination is made to route the traffic via the secondary network, a first address (e.g., an IP address) associated with a communication device (e.g., a client device) may be translated/mapped to a secondary address in block **216***b*. Such a translation/mapping may facilitate a conveyance/transmission of the traffic in an upward/uplink direction. Conversely, traffic in a downward/downlink direction in the secondary network may undergo translation from a/the secondary address to the first address as part of block **216***b*, to facilitate providing the traffic to the communication device. If the traffic is determined/identified as being routed via the primary network as part of block **212***b*, then address translation might not be used; e.g., the first address may be used/selected as part of block **216***b*.

[0042] In block **220***b*, the traffic may be routed from a source device to one or more destination devices based on the address selected as part of block **216***b*. In this manner, the one or more destination devices may operate upon the traffic, potentially as part of an execution of one or more applications. Block **220***b* may include generating/establishing one or more networks, such as for example one or more secondary networks as described above. To the extent that such secondary networks already exist, block **220***b* may include utilizing such secondary networks.

[0043] A generation or establishment of one or more networks (such as, for example, as part of block **220***b*) may include a placement, deployment or allocation of physical resources to support the network(s). In some embodiments, the generation or establishment of one or more networks (such as, for example, as part of block **220***b*) may include a generation of a virtual and/or private network utilizing (pre) existing physical resources.

[0044] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. **2**B, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0045] Aspects of this disclosure are directed to systems, apparatuses, and methods for creating/generating/establishing networks (e.g., private networks) on demand to host gaming events like player-vs-player matches, tournaments, leagues or other high-profile events. A game developer (or other entity/party) may be provided with an ability to establish a private network between game servers and any participants in the event, preventing bad actors from identifying where the game servers exist, injecting malicious traffic into the private network, or impacting the other players in the event. In some embodiments, the private network may be established or may extend between a gateway or client device of each player and a game server. For users on a mobility network, a network element may be dynamically inserted into a data plane that would achieve the same, or similar, functionality. In some embodiments, a use of an OTT client application may enable various users (such as users that are not subscribed to a service supported by a gaming architecture/platform) to connect to the private network as well. Aspects of this disclosure could also potentially make use of/leverage unused bandwidth between, e.g., a gateway and a network operator/service provider backbone network when subsidized by a game developer/operator, for example.

[0046] As set forth herein, aspects of this disclosure may be included/incorporated/integrated as part of one or more practical applications. For example, aspects of this disclosure may ensure that information associated with a user or communication device is not compromised by shielding or obscuring that information from view. In this manner, the user and the communication device might not be subjected to attacks by nefarious actors that may desire to impose harm. Further, aspects of this disclosure may ensure a fair/level playing field as part of a competition (e.g., an online game) via the use of a separate/secondary network as part of the competition.

[0047] Aspects of this disclosure represent substantial improvements to the state of the art. For example, security of information associated with a user (or, analogously, a device) may be enhanced without incurring any penalty in terms of latency or a degraded quality of service (QOS) or quality of experience (QoE). In fact, aspects of this disclosure may enhance QoS and QoE via an improvement or enhancement in a routing of traffic by reducing a number of devices or hops between a source device and a target/intended destination device. This reduction in number of hops further reduces the likelihood/probability that information may be compromised.

[0048] Aspects of this disclosure may include an imposition or application of address translation in respect of one or more connections associated with one or more communication devices. To the extent that a given communication device supports one or more technologies, such as one or more radio access technologies (RATs), respective translations may be imposed on each of the RATs or associated connections. Thus, a device that is operative in conjunction with 5G and LTE technologies may have a first translation imposed in respect of a 5G connection and a second translation imposed in respect of an LTE connection; the second translation may be the same as, or different from, the first translation.

[0049] As described herein, a primary network may be used to convey (or operate upon) first traffic/data and a secondary network may be used to convey (or operate upon) second traffic/data.

From the perspective of an end-user or client device, the presence of the secondary network may be obscured or shielded from view, which is to say that it may appear to the end-user or client device that the second traffic/data is being conveyed via the primary network. In this manner, the end-user/client device might not be burdened with the details pertaining to the routing or conveyance of traffic/data. Further, by placing/locating at least some of the functionality described herein outside of the purview/domain of a client device, client device resources (such as, for example: battery life/capacity, memory resources, etc.) may be preserved. Additionally, aspects of this disclosure may reduce the amount of time or overhead associated with a development and maintenance of an application (e.g., game), thereby alleviating a developer/maintainer of the application of the costs associated with such activities.

[0050] As described above, aspects of this disclosure may include a placement or distribution of functionality at one or more locations and/or at one or more communication devices. In some embodiments, a network may be established to facilitate an application, a competition, a game, or the like. The network may be dynamically extended to one or more locations, such as for example via a gateway, a phone, a hotspot, etc. In some embodiments, a virtualized or containerized network function hosted on, e.g., a gateway or dynamically inserted into a mobility user/data plane identifies a user's/subscriber's gaming traffic and separates that traffic from typical Internet-bound (non-gaming) traffic, routing the identified packets through the established network.

[0051] In some embodiments, one or more networks may be established or utilized to facilitate a conveyance or routing of traffic/data. In some embodiments, multiple secondary networks may be generated/utilized. The networks that are used in some embodiments may include physical or virtual networks. In some embodiments, competitions may be routed through a particular network. In some embodiments, participant devices in a particular competition may be intentionally/selectively routed through different ones of a plurality of secondary networks for added security, for distributing load, etc.

[0052] In some embodiments, network related information (including, for example, address information used to facilitate a translation or mapping) may be stored at one or more devices. At least some of the information may be deleted upon use, at the conclusion of a session (e.g., at the conclusion of an event or competition), etc., to promote security/privacy. In some instances, the information may be communicated from a first network element or device to other network elements or devices, such as for example when multiple secondary networks are utilized, in order to maintain persistency/continuity of a session from one network to the next. Even in such instances, the information may be shared/distributed (only) to the extent necessary, in order to promote/maintain security.

[0053] Aspects of this disclosure may reduce the probability of a DDOS or other attack. Still further, aspects of this disclosure may disincentivize cheating in relation to a game or competition by preventing outside traffic from reaching peer connections or a game server and by tying network identity to a real subscriber or user identity. As set forth herein, aspects of this disclosure may be applied in respect of wireline/wired technology and wireless technology, thereby finding a wide range of applicability in the networking community.

[0054] Referring now to FIG. **3**, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, the subsystems and functions of system **200***a*, and method **200***b* presented in FIGS. **1**, **2**A, and **2**B. For example, virtualized communication network **300** can facilitate in whole or in part monitoring first traffic from a first device, wherein the first device has a first network address associated with a first network, classifying, based on the monitoring of the first traffic, the first traffic as first gaming traffic, based on the classifying of the first traffic as first gaming traffic, establishing a second network that is different from the first network, mapping the first network address to a second network address that is different from the first network address, wherein the second network address is associated with the second network, and routing the first traffic to a gaming server of the second network using the second network address. Virtualized communication network **300** can facilitate in whole or in part obtaining first traffic from a first communication device, analyzing the first traffic to classify the first traffic as being associated with an execution of a first application, based on the classification of the first traffic as being associated with the execution of the first application, translating an address associated with the first traffic from a first address associated with a first network to a second address associated with a second network, the second network being different from the first network, and conveying the first traffic to a second communication device of the second network using the second address. Virtualized communication network **300** can facilitate in whole or in part analyzing first data to identify that the first data is associated with an online game and to identify that the first data is associated with a first device, the first device having a first address associated with a first network, establishing, based on the analyzing of the first data, a second network that is different from the first network, translating the first address to a second address that is different from the first address, wherein the second address is associated with the second network, and transmitting the first data to a communication device in the second network using the second address.

[0055] In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

[0056] In contrast to traditional network elements-which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general purpose processors or general purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

[0057] As an example, a traditional network element **150** (shown in FIG. **1**), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it's elastic: so the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle-boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

[0058] In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized, and might require special DSP code and analog front-ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

[0059] The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributers and other network elements. Because these elements don't typically need to forward large amounts of traffic, their workload can be distributed across a number of servers—each of which adds a portion of the capability, and overall which creates an elastic function with higher availability than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

[0060] The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330**, **332**, **334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular, network workloads may have applications distributed across the virtualized network function cloud **325** and cloud computing environment **375** and in the commercial cloud, or might simply orchestrate workloads supported entirely in NFV infrastructure from these third party locations.

[0061] Turning now to FIG. **4**, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. **4** and the following discussion are intended to provide a brief, general description of a suitable computing environment **400** in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment **400** can be used in the implementation of network elements **150**, **152**, **154**, **156**, access terminal **112**, base station or access point **122**, switching device **132**, media terminal **142**, and/or VNEs **330**, **332**, **334**, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment **400** can facilitate in whole or in part monitoring first traffic from a first device, wherein the first device has a first network address associated with a first network, classifying, based on the monitoring of the first traffic, the first traffic as first gaming traffic, based on the classifying of the first traffic as first gaming traffic, establishing a second network that is different from the first network, mapping the first network address to a second network address that is different from the first network address, wherein the second network address is associated with the second network, and routing the first traffic to a gaming server of the second network using the second network address. Computing environment **400** can facilitate in whole or in part obtaining first traffic from a first communication device, analyzing the first traffic to classify the first traffic as being associated with an execution of a first application, based on the classification of the first traffic as being associated with the execution of the first application, translating an address associated with the first traffic from a first address associated with a first network to a second address associated with a second network, the second network being different from the first network, and conveying the first traffic to a second communication device of the second network using the second address. Computing environment **400** can facilitate in whole or in part analyzing first data to identify that the first data is associated with an online game and to identify that the first data is associated with a first device, the first device having a first address associated with a first network, establishing, based on the analyzing of the first data, a second network that is different from the first network, translating the first address to a second address that is different from the first address, wherein the second address is associated with the second network, and transmitting the first data to a communication device in the second network using the second address.

[0062] Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0063] As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

[0064] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0065] Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

[0066] Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0067] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0068] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0069] With reference again to FIG. **4**, the example environment can comprise a computer **402**, the computer **402** comprising a processing unit **404**, a system memory **406** and a system bus **408**. The system bus **408** couples system components including, but not limited to, the system memory **406** to the processing unit **404**. The processing unit **404** can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit **404**.

[0070] The system bus **408** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **406** comprises ROM **410** and RAM **412**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **402**, such as during startup. The RAM **412** can also comprise a high-speed RAM such as static RAM for caching data.

[0071] The computer **402** further comprises an internal hard disk drive (HDD) **414** (e.g., EIDE, SATA), which internal HDD **414** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **416**, (e.g., to read from or write to a removable diskette **418**) and an optical disk drive **420**, (e.g., reading a CD-ROM disk **422** or, to read from or write to other high capacity optical media such as the DVD). The HDD **414**, magnetic FDD **416** and optical disk drive **420** can be connected to the system bus **408** by a hard disk drive interface **424**, a magnetic disk drive interface **426** and an optical drive interface **428**, respectively. The hard disk drive interface **424** for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0072] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **402**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0073] A number of program modules can be stored in the drives and RAM **412**, comprising an operating system **430**, one or more application programs **432**, other program modules **434** and program data **436**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **412**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0074] A user can enter commands and information into the computer **402** through one or more wired/wireless input devices, e.g., a keyboard **438** and a pointing device, such as a mouse **440**. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit **404** through an input device interface **442** that can be coupled to the system bus **408**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

[0075] A monitor **444** or other type of display device can be also connected to the system bus **408** via an interface, such as a video adapter **446**. It will also be appreciated that in alternative embodiments, a monitor **444** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **402** via any communication means, including via the Internet and cloud-based networks. In addition to the monitor **444**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

[0076] The computer **402** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **448**. The remote computer(s) **448** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer **402**, although, for purposes of brevity, only a remote memory/storage device **450** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **452** and/or larger networks, e.g., a wide area network (WAN) **454**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0077] When used in a LAN networking environment, the computer **402** can be connected to the LAN **452** through a wired and/or wireless communication network interface or adapter **456**. The adapter **456** can facilitate wired or wireless communication to the LAN **452**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **456**.

[0078] When used in a WAN networking environment, the computer **402** can comprise a modem **458** or can be connected to a communications server on the WAN **454** or has other means for establishing communications over the WAN **454**, such as by way of the Internet. The modem **458**, which can be internal or external and a wired or wireless device, can be connected to the system bus **408** via the input device interface **442**. In a networked environment, program modules depicted relative to the computer **402** or portions thereof, can be stored in the remote memory/storage device **450**. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0079] The computer **402** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0080] Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

[0081] Turning now to FIG. **5**, an embodiment **500** of a mobile network platform **510** is shown that is an example of network elements **150**, **152**, **154**, **156**, and/or VNEs **330**, **332**, **334**, etc. For example, platform **510** can facilitate in whole or in part monitoring first traffic from a first device, wherein the first device has a first network address associated with a first network, classifying, based on the monitoring of the first traffic, the first traffic as first gaming traffic, based on the classifying of the first traffic as first gaming traffic, establishing a second network that is different from the first network, mapping the first network address to a second network address that is different from the first network address, wherein the second network address is associated with the second network, and routing the first traffic to a gaming server of the second network using the second network address. Platform **510** can facilitate in whole or in part obtaining first traffic from a first communication device, analyzing the first traffic to classify the first traffic as being associated with an execution of a first application, based on the classification of the first traffic as being associated with the execution of the first application, translating an address associated with the first traffic from a first address associated with a first network to a second address associated with a second network, the second network being different from the first network, and conveying the first traffic to a second communication device of the second network using the second address. Platform **510** can facilitate in whole or in part analyzing first data to identify that the first data is associated with an online game and to identify that the first data is associated with a first device, the first device having a first address associated with a first network, establishing, based on the analyzing of the first data, a second network that is different from the first network, translating the first address to a second address that is different from the first address, wherein the second address is associated with the second network, and transmitting the first data to a communication device in the second network using the second address.

[0082] In one or more embodiments, the mobile network platform **510** can generate and receive signals transmitted and received by base stations or access points such as base station or access point **122**. Generally, mobile network platform **510** can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform **510** can be included in telecommunications carrier networks, and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform **510** comprises CS gateway node(s) **512** which can interface CS traffic received from legacy networks like telephony network(s) **540** (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network **560**. CS gateway node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology(ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

[0083] In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

[0084] In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

[0085] For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In addition to application server, server(s) **514** can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550** or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. **1**(*s*) that enhance wireless service coverage by providing more network coverage.

[0086] It is to be noted that server(s) **514** can comprise one or more processors configured to confer at least in part the functionality of mobile network platform **510**. To that end, the one or more processor can execute code instructions stored in memory **530**, for example. It is should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

[0087] In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

[0088] In order to provide a context for the various aspects of the disclosed subject matter, FIG. **5**, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

[0089] Turning now to FIG. **6**, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, computing device **600** can facilitate in whole or in part monitoring first traffic from a first device, wherein the first device has a first network address associated with a first network, classifying, based on the monitoring of the first traffic, the first traffic as first gaming traffic, based on the classifying of the first traffic as first gaming traffic, establishing a second network that is different from the first network, mapping the first network address to a second network address that is different from the first network address, wherein the second network address is associated with the second network, and routing the first traffic to a gaming server of the second network using the second network address. Computing device **600** can facilitate in whole or in part obtaining first traffic from a first communication device, analyzing the first traffic to classify the first traffic as being associated with an execution of a first application, based on the classification of the first traffic as being associated with the execution of the first application, translating an address associated with the first traffic from a first address associated with a first network to a second address associated with a second network, the second network being different from the first network, and conveying the first traffic to a second communication device of the second network using the second address.

[0090] Computing device **600** can facilitate in whole or in part analyzing first data to identify that the first data is associated with an online game and to identify that the first data is associated with a first device, the first device having a first address associated with a first network, establishing, based on the analyzing of the first data, a second network that is different from the first network, translating the first address to a second address that is different from the first address, wherein the second address is associated with the second network, and transmitting the first data to a communication device in the second network using the second address.

[0091] The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a user interface (UI) **604**, a power supply **614**, a location receiver **616**, a motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, WiFi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-**1**X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VoIP, etc.), and combinations thereof.

[0092] The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing assembly of the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

[0093] The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.
[0094] The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.
[0095] The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.
[0096] The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).
[0097] The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, WiFi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.
[0098] Other components not shown in FIG. **6** can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.
[0099] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and doesn't otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0100] In the subject specification, terms such as “store,” “storage,” “data store,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, nonvolatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.
[0101] Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.
[0102] In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.
[0103] Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically identifying acquired cell sites that provide a maximum value/benefit after addition to an existing communication network) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site of the acquired network. A classifier is a function that maps an input attribute vector, x=(x1, x2, x3, x4, . . . , xn), to a confidence that the input belongs to a class, that is, f(x)=confidence (class). Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic

models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

[0104] As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria which of the acquired cell sites will benefit a maximum number of subscribers and/or which of the acquired cell sites will add minimum value to the existing communication network coverage, etc.

[0105] As used in some contexts in this application, in some embodiments, the terms "component," "system" and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0106] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term "article of manufacture" as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0107] In addition, the words "example" and "exemplary" are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as "example" or "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0108] Moreover, terms such as “user equipment,” “mobile station,” “mobile,” subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

[0109] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0110] As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

[0111] As used herein, terms such as “data storage,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

[0112] What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0113] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0114] As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

[0115] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

Claims

1. A device, comprising: a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: monitoring first traffic from a first device, wherein the first device has a first network address associated with a first network, wherein the monitoring is based on an analysis of a header of the first traffic and metadata of the first traffic, and wherein the monitoring does not access a payload of the first traffic; classifying, based on the monitoring of the first traffic, the first traffic as being associated with a first application; based on the classifying of the first traffic as being associated with the first application, translating the first network address to a second network address that is different from the first network address, wherein the second network address is associated with a second network; and routing the first traffic to a second device of the second network using the second network address, wherein the second device is excluded from the first network.

2. The device of claim 1, wherein the first application is a first gaming application.

3. The device of claim 1, wherein the operations further comprise: monitoring second traffic from the first device; classifying, based on the monitoring of the second traffic, the second traffic as

being associated with a second application that is different from the first application; and based on the classifying of the second traffic as being associated with the second application, routing the second traffic using the first network address.
4. The device of claim 1, wherein the first network address includes an Internet Protocol (IP) address.
5. The device of claim 1, wherein the first device is a gateway.
6. The device of claim 1, wherein the first device is a client device.
7. The device of claim 1, wherein the operations further comprise: monitoring second traffic; identifying, based on the monitoring of the second traffic, the second traffic as having the second network address associated therewith; based on the identifying, mapping the second network address to the first network address; and routing, based on the mapping of the second network address to the first network address, the second traffic to the first device.
8. The device of claim 1, wherein the operations further comprise: monitoring second traffic from a third device, wherein the third device has a third network address associated with the first network, and wherein the third network address is different from the first network address; classifying, based on the monitoring of the second traffic, the second traffic as being associated with the first application; based on the classifying of the second traffic as being associated with the first application, mapping the third network address to a fourth network address that is different from the third network address, wherein the fourth network address is associated with the second network; and routing the second traffic using the fourth network address.
9. The device of claim 8, wherein the operations further comprise: monitoring third traffic; identifying, based on the monitoring of the third traffic, the third traffic as having the fourth network address associated therewith; based on the identifying of the third traffic as having the fourth network address associated therewith, mapping the fourth network address to the third network address; and routing, based on the mapping of the fourth network address to the third network address, the third traffic to the third device.
10. The device of claim 8, wherein the first device and the third device are included as part of a team in a gaming competition.
11. The device of claim 8, wherein the first device is included as part of a first team in a gaming competition, wherein the third device is included as part of a second team in the gaming competition, and wherein the second team is different from the first team.
12. A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising: monitoring first traffic from a first device, wherein the first device has a first network address associated with a first network, wherein the monitoring is based on an analysis of a header of the first traffic, metadata of the first traffic, or a combination thereof, and wherein the monitoring does not access a payload of the first traffic; classifying, based on the monitoring of the first traffic, the first traffic as being associated with a first gaming application; based on the classifying of the first traffic as being associated with the first gaming application, translating the first network address to a second network address that is different from the first network address, wherein the second network address is associated with a second network; and routing the first traffic to a second device of the second network using the second network address, wherein the second device is excluded from the first network.
13. The non-transitory machine-readable medium of claim 12, wherein the first traffic includes downlink traffic.
14. The non-transitory machine-readable medium of claim 12, wherein the first traffic includes uplink traffic.
15. The non-transitory machine-readable medium of claim 12, wherein the operations further comprise: obtaining second traffic from the first device; analyzing the second traffic to classify the second traffic as being associated with an execution of a second application, wherein the second

application is different from the first gaming application; and based on the classification of the second traffic as being associated with the execution of the second application, conveying the second traffic to a third device using the first network address.

16. The non-transitory machine-readable medium of claim 12, wherein the operations further comprise: obtaining second traffic from the first device; and analyzing the second traffic to classify the second traffic as being associated with an execution of the first gaming application.

17. The non-transitory machine-readable medium of claim 16, wherein the operations further comprise: based on the classification of the second traffic as being associated with the execution of the first gaming application, translating an address associated with the second traffic from the first network address to a third network address, wherein the third network address is different from the second network address, and wherein the third network address is associated with the second network; and conveying the second traffic using the third network address.

18. A method, comprising: monitoring, by a processing system including a processor, first traffic from a first device, wherein the first device has a first network address associated with a first network, wherein the monitoring is based on an analysis of a header of the first traffic, metadata of the first traffic, or a combination thereof, and wherein the monitoring does not access a payload of the first traffic; classifying, by the processing system and based on the monitoring of the first traffic, the first traffic as being associated with a first gaming application; based on the classifying of the first traffic as being associated with the first gaming application, translating, by the processing system, the first network address to a second network address that is different from the first network address, wherein the second network address is associated with a second network; and routing, by the processing system, the first traffic to a second device using the second network address, wherein the second device is excluded from the first network.

19. The method of claim 18, wherein the first network is a public network and the second network is a private network.

20. The method of claim 19, further comprising: transmitting, by the processing system, second traffic to the second device as part of a third network using a third network address, the third network including a virtual network.